

Practice Quiz 02A — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 02

Conditionals, contrapositives, and free and bound variables.

1. (a) Write the converse and contrapositive of $\neg P \rightarrow Q$. Make one complete four-row truth table comparing all three formulas, and state which of the two new formulas is equivalent to the original.

Solution The converse is $Q \rightarrow \neg P$. The contrapositive is $\neg Q \rightarrow \neg\neg P$, equivalently $\neg Q \rightarrow P$.

P	Q	$\neg P \rightarrow Q$	$Q \rightarrow \neg P$	$\neg Q \rightarrow P$
F	F	F	T	F
F	T	T	T	T
T	F	T	T	T
T	T	T	F	T

The contrapositive has the same truth values as the original in all four rows. The converse does not: for example, the first row differs. ■

- (b) In $\exists t \in \mathbf{R} (t^2 + c = w)$, identify the free and bound variables.

Solution The variable t is bound by $\exists t$. The variables c and w are free. Being inside the quantifier's scope does not bind c or w . ■